

Reproducing “Bilinear MLPs Enable Weight-Based Mechanistic Interpretability”

FACT-AI Course Project

January 28, 2026

Bilinear MLPs: Weight-Based Interpretability

Key Innovation:

- ▶ Direct interpretability from weights
- ▶ No post-hoc analysis needed
- ▶ No SAE training required

Architecture:

$$\mathbf{y} = (\mathbf{W}_l \mathbf{x}) \odot (\mathbf{W}_r \mathbf{x})$$

Element-wise multiplication creates quadratic interactions

Motivation:

- ▶ Standard: train SAEs on activations
- ▶ Bilinear: interpret weights directly
- ▶ Interaction tensor eigendecomposable
- ▶ Reveals features without auxiliary models

Goal:

- ▶ Understand decisions through weight structure
- ▶ Enable transparent AI systems

Eigendecomposition Methodology

Method:

1. Extract interaction tensor: $B_c = \mathbf{W}_l \otimes \mathbf{W}_r$
2. Symmetrize: $B_c^{\text{sym}} = \frac{1}{2}(B_c + B_c^\top)$
3. Eigendecomposition:

$$B_c^{\text{sym}} = \sum_{r=1}^d \lambda_r^{(c)} \mathbf{v}_r^{(c)} (\mathbf{v}_r^{(c)})^\top$$

Effective Rank:

$$\text{EffRank}(\boldsymbol{\lambda}) = \left(\frac{\|\boldsymbol{\lambda}\|_1}{\|\boldsymbol{\lambda}\|_2} \right)^2$$

Low effective rank $\Rightarrow$ few dominant eigenvectors $\Rightarrow$ interpretable structure

Vision Results: Regularization Induces Low-Rank Structure

Key Finding:

- ▶ Effective rank: **15.5** (WD) vs **38.5** (none)
- ▶ Ratio: **0.40** < 0.5 ✓
- ▶ Accuracy: **97–98%**

Main Driver:

- ▶ Weight decay drives low-rank structure
- ▶ Noise improves accuracy but increases rank
- ▶ WD alone achieves lowest effective rank

Claim V1: Fully reproduced ✓

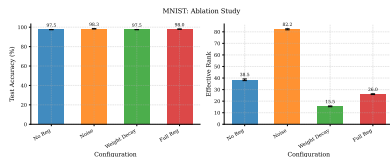


Figure: Accuracy vs. effective rank

Vision Results: Interpretable Eigenvectors

MNIST (Full Regularization: $\sigma=0.5$, $\lambda=1.0$): Top Eigenvectors

Regularized Models:

- Clear digit patterns
- Recognizable stroke templates
- Interpretable geometric features

Unregularized Models:

- Diffuse, noise-like patterns
- No clear structure
- Overfitting to training data

Claims V2 & V3: Fully reproduced



Figure: Regularized eigenvectors (top 10 per class)

Language Results: Negation Circuit Discovery

AND-Gate Circuit:

- ▶ Feature 751: strongest AND-gate
- ▶ AND-score: **0.96**
- ▶ Eigenvalues: $\lambda_1 = -0.46$, $\lambda_2 = +0.18$

Semantic Structure:

- ▶ **Positive:** “wonderful”, “excellent”, “fortunate”
- ▶ **Negative:** “horrible”, “disease”, “unfair”
- ▶ Clear block structure

Claims L1 & L2: Fully reproduced

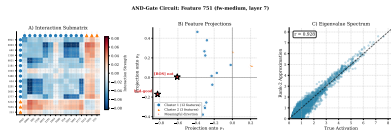


Figure: Negation circuit: Feature 751

Language Results: Correlation Gap & Root Cause

Claim L3: 69% achieve >0.75
rank-2 correlation

Our Result:

- ▶ Only **18–32%** achieve >0.75
- ▶ Mean correlation: 0.27–0.51
- ▶ Significant gap from claim

Root Cause:

- ▶ SAE training duration
- ▶ Correlation improves **2.5×** with longer training
- ▶ Public SAEs less trained than paper's internal checkpoints

Methodology sound; gap from SAE quality

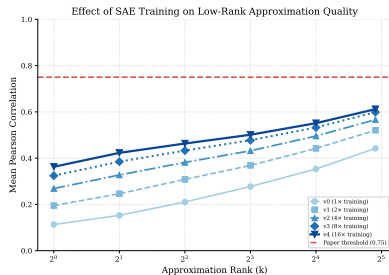


Figure: Correlation progression with SAE training

Extension 1: Cross-Dataset Robustness

Hypothesis:

- ▶ Models learn **geometric circuits**
- ▶ Eigenvectors = “Platonic forms” (circularity)

Test Cases:

- ▶ Writer Independence: MNIST $\leftrightarrow$ EMNIST-Digits
- ▶ Domain Transfer: MNIST $\rightarrow$ USPS
- ▶ Geometric Semantics: MNIST $\rightarrow$ EMNIST-Letters

Results:

- ▶ MNIST $\rightarrow$ EMNIST: **97.7%**
- ▶ MNIST $\rightarrow$ USPS: **72.3%** (+19 pp)
- ▶ Letters: O $\rightarrow$ 0 (99.6%), I $\rightarrow$ 1 (87.1%), Z $\rightarrow$ 2 (88.8%), S $\rightarrow$ 5

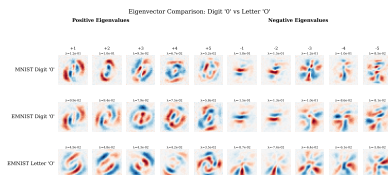


Figure: Three-way eigenvector comparison

Quadratic Form Similarity

Problem: Cosine Similarity Fails

- ▶ Similar (O-O): **0.358**
- ▶ Dissimilar (O-X): **0.339**
- ▶ Not discriminative!

Root Cause:

- ▶ Shared eigenvector directions
- ▶ Class info in eigenvalue *weights*
- ▶ Cosine ignores magnitudes

Solution:

$$\text{sim}(A, B) = \frac{\text{tr}(A \cdot B)}{\|A\|_F \|B\|_F}$$

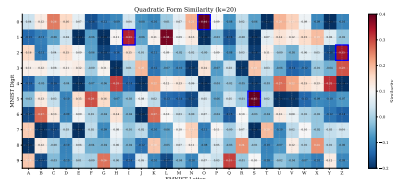


Figure: Quadratic Form Similarity heatmap

Results:

- ▶ Similar: **0.400**
- ▶ Dissimilar: **-0.060**
- ▶ Gap: $p < 10^{-4}$

Incorporates eigenvalue signs and alignment

Extension 2: CP-Decomposition

Motivation:

- ▶ Post-hoc → **intrinsic** interpretability
- ▶ Standard eigendecomposition: orthogonality → entangled
- ▶ CP: disentangled features

CP Factorization:

$$W_{cij} = \sum_{r=1}^R \lambda_r A_{cr} B_{ir} C_{jr}$$

Key Properties:

- ▶ No orthogonality constraint
- ▶ Disentangled representations
- ▶ Parameter reduction:
 $\mathcal{O}(R \times d_{in})$

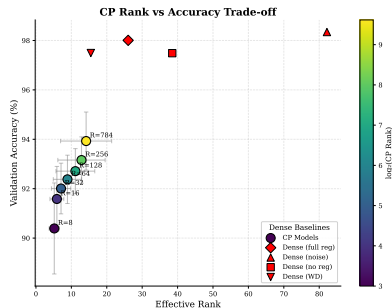


Figure: Accuracy vs rank trade-off

Results at $R = 256$:

- ▶ Accuracy: **93.8%**
- ▶ Effective rank: **17.5**
- ▶ vs Dense: 97.5% (3.7 pp drop)

CP-Decomposition: Disentangled Factors

Dense Baseline (Full Reg)

Top 5 Eigenvectors Comparison: Dense vs CP

CP Model (R=64, lambda)

